Topic : Compiler Construction

Chapter 1 : Introduction to Compiler

1.1: What is Compiler?

1.2: History & Evolution of compilers

1.3: Language Processors

1.4 : Compiler vs Interpreter

1.5: Structure of a Compiler

1.6 : Phases of Compiler

1.7: Front End & Back End of Compiler

1.8: Passes of Compiler (Single pass, Multi pass)

1.9 : Bootstrapping of Compiler

1.10: Challenges in Compiler Design

1.11:Modern compiler Examples

1.12: Application of compiler Technolgy

1.13: Error Handling in Compiler

Chapter 2 : Lexical Analysis

2.1: Role of Lexical Analyzer

2.2 : Tokens, Lexemes, Patterns

2.3 : Specification of Tokens

2.4 : Regular Expressions

4.12: Attribute Grammars

4.13: Syntax Directed Translations (SDT)

4.14: Semantic Error Handling

4.15: Intermediate Representation (Introduction)

4.16: Abstract Syntax Tree (AST)

Chapter 5 : Intermediate Code Generation

5.1: Need for Intermediate Code

5.2: Role of IR in Compiler

5.3: Types of Intermediate Representation

5.4: Three Address Code (TAC)

5.5: Representation of TAC

5.6: Syntax-Directed Translation for ICG

5.7:Translation of Expressions

5.8: Translation of Boolean Expressions

5.9: Short-Circuit Evaluation

5.10: Translation of Statements

5.11: Control Flow Statements

5.12: Backpatching

5.13:Abstract Syntax Tree to IR Conversion

5.14: DAG Representation for Expression

Chapter 6 : Code Optimization

6.1: Introduction to Code Optimization

6.2: Need for Optimization

Chapter 7 : Code Generation

7.8: Code Generation for Expressions

7.9: Code Generation for Boolean Expression

7.10: Code Generation for Control statement

7.11: Code Generation for Procedure Calls

7.12: Calling Conventions

7.13: Stack frame & Activation Record Layout

7.14: Runtime Storage Management

7.15: Instruction Scheduling

7.16: Spill code handling

7.17:Error handling and debugging support

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8.1: Introduction to Runtime Environment

8.2 :Source Language Issues

8.3: Binding Time

8.4:Storage Organization

8.5 :Storage Allocation Strategies

8.6: Procedure Call and Return Mechanism

8.7 :Scope Rules & Accessing Non-local Variables

8.8: Storage Allocation Strategies

8.9: Dynamic Storage Management

8.10 :Runtime Support for Control Flow

8.11: Runtime Environment for Object-Orianted languages

8.12 :Error Handling at Runtime

8.13 : Garbage Collection Techniques

Chapter 9 : Error Handling

Chapter 10: Modern Compiler Architecture and Industry practices

10.12: Cross Compilation & Embedded Systems

10.13: LLVM & Modern compiler frameworks

10.14: Cloud & Distributed Compilation

10.15: AI & ML Compiler Concepts